

Rafael Formes

Game Developer

Philippines

Portfolio | rafformes@gmail.com | <https://www.linkedin.com/in/rafformes/> | <https://github.com/leafarz>

Skills

Programming Languages: C++, C#, Python, Typescript
Game Engines: UE4, UE5, Unity, Cocos2d-x
3D: OpenGL, DirectX12, GLSL, three.js, AFrame, Phaser
Other Tools / Libraries: .NET, Firebase, NestJs, Jenkins, Git, SVN, Perforce

Professional Experience (2013-Present)

● Game Developer

(May 2025 - Present)

UE5 | C++ | Common UI | GAS | State Trees | VFX

○ Phobos (PC)

- Isometric action RPG that my colleague and I are working on as a side project. Game is still early in development.
- Developed with UE5.6 and fully utilizing GAS, motion matching, state trees and gameplay camera system
- Developed the core game framework, enemy AI, inventory & equipment system, loadouts, skill tree system, UI and VFX using Niagara system.
- Links
 - Recording 1: <https://drive.google.com/file/d/191Nsk3KKmnAf9e1UR9dZ5gASQFzNppAg/view>
 - Recording 2: https://drive.google.com/file/d/17L2o7odKTYdhTt_51KpIDxWRh_iDXKix/view

● Game Developer at Eternal Squares, UK

(May 2025 - Sept 2025)

UE5 | C++ | Common UI | GAS | Multiplayer | Perforce | UGS | ImmutableX | Blockchain | Typescript | AWS | GCP

○ Eternal League (PC)

- Work on a project in a team of 2 developers.
- Integrate blockchain technology using Immutable X and work on bug fixes.
- Implement and deploy backend game apis, blockchain functionalities and frontend shop.
- Links
 - Epic Games: <https://store.epicgames.com/en-US/p/eternal-league-5e5031>
 - Steam: https://store.steampowered.com/app/3542920/Eternal_League/

- **Game Developer** at Mon Co. (LiquidX Studio), SG (Aug 2024 - Apr 2025)
 UE5 | C++ | Common UI | Gameplay Ability System / GAS | Multiplayer | Perforce | UGS
 - Eternal League (PC, Android)
 - Worked on a project in a team of developers, artists, vfx, designers, etc.
 - Implement, optimize and maintain game mechanics, UI, tools in UE5 and documentations.
 - Implemented fog of war, skill tree, authentication, campaign progression, tutorial, game modifiers, game consumables, etc.
- **Lead Software Developer** at Touchcast Inc., NY (Jun 2022 - Aug 2024)
 UE4 | UE5 | C++ | C# | Python | Jenkins
 - Touchcast Metaverse Studio (PC)
 - Additional responsibilities: Lead a team of software engineers, design code architectures, and plan technical requirements.
- **Senior Unreal Engine Developer** at Touchcast Inc., NY (Mar 2021 - Jun 2022)
 UE4 | UE5 | C++ | C# | Python | Jenkins
 - Touchcast Metaverse Studio (PC)
 - Implement features, Widgets / UI, materials, editor tools and APIs in UE4 and UE5.
 - Developed the system for downloading of environments and loading at runtime.
 - Developed the build pipeline using Jenkins, C# and Python.
 - Profiling and optimization of code, materials and environments.
- **.NET Developer** at UCIT, Philippines (Stealth Monitoring, Inc.), PH (Nov 2019 - Apr 2021)
 .NET C# | Azure
 - Build security and surveillance applications using various languages and platforms
 - Code reviews, implement user stories and develop microservices.
- **Senior Game Developer** at The Studio Of Secret 6 Inc., PH (Aug 2017 - Nov 2019)
 UE4 | Unity | Javascript | three.js | Firebase | Phaser
 - MediEvil Remake for (PS4, ported from PS1)
 - Joined mid development using UE4 with a team of 3 programmers.
 - Dash mechanic in CharacterMovement
 - Auto-walk to waypoint interaction
 - Push and pull mechanics for rotary and multidirectional object interactions
 - Optimization of physics, transparencies and animation culling.
 - Link: <https://www.playstation.com/en-us/games/medieval/>
 - Giant Cop for (PS4 VR)
 - Worked on optimizations for using Unity3d with a team of 3 programmers
 - Worked on animation culling, LODs and frame rate prioritization of objects
 - Code refactoring and spreading of the Update function timings.
 - Link: https://store.steampowered.com/app/451080/Giant_Cop_Justice_Above_All/

- Poker game in AltspaceVR
 - Developed and shipped a game with a team of 2 programmers
 - Used three.js library and Firebase for network gameplay
 - I implemented the custom Firebase library which was used for multiple AltspaceVR games that use network features.
- Adventure Time: Blind Finned & Bugs Bunny: Carrot Crisis (Web app)
 - Developed and shipped using Phaser from the ground up.
 - Link 1: <https://www.cartoonnetwork.co.uk/games/adventure-time-blind-finned>
 - Link 2: <https://www.cartoonnetworkasia.com/games/looney-tunes-cartoons-carrot-crisis/play>
- **Game Developer** at The Studio Of Secret 6 Inc., PH (Apr 2014 - Jan 2017)
 Unity | C# | Javascript | three.js | Firebase | Cocos-2d
 - Love Letter, Boss Monsters and Dungeons & Dragons (three.js | AltspaceVR)
 - Develop and ship using three.js from the ground up with a team of 2 programmers.
 - I worked on the gameplay logic and the resource management.
 - Develop our network library using Firebase.
 - YuGiOh Duel Generation (Cocos-2d | Android and iOS) - Developed and did bug fixes.
 - Broadsword: Age of Chivalry v2 (Unity | Android and iOS) - Developed and did bug fixes.
 - Star Wars Science: The Force™ Trainer II Hologram Experience (Unity | Android and iOS)
 - Developed and shipped using Unity
 - I worked on the lighting vfx and the lightsaber assembly via math computations.
 - Tinted (Unity | Android and iOS)
 - I worked on the match-3 system from the ground up. Implemented the effect of different paint drops mixing in the background.
- **Game Programmer** at Other Ocean Interactive, NL, Canada (Feb 2017 - Jun 2017)
 Unity | C# | Elixir
 - Optimize and contribute to bug fixes for Reload: The Z-Team using Unity and implement backend APIs using Elixir.
- **Software Engineer Intern** at KLAB Cyscorpions, Inc., PH (Apr 2013 - May 2013)
 Unity | C#
 - Develop Grav in collaboration with KLAB for Android and iOS using Unity. Game is from our game jam we submitted around 2012.

Personal Projects

- **ueSandbox** at <https://github.com/leafarz/ueSandbox> (Apr 2022 - Present)
 - Sandbox project for anything programming / UE related.
 - Includes a demo inspired from the game The Unfinished Swan.
 - Other examples include utilizing render targets, materials and post processes.
- **Pagong** at <https://github.com/leafarz/Pagong> (Dec 2023)
 - Game engine using DirectX12.
 - It's currently at its early stages but the goal at least is to have a model and shader viewer.
- **gg** at <https://github.com/leafarz/gg> (Dec 2017 - Jul 2018)
 - Game engine using OpenGL.
 - No plans of continuing at the moment but this can render models, has lighting, shaders and render textures. The UI library used is ImGui.

Education

- **University of the Philippines Diliman** (Jun 2010 - Apr 2014)
 - BS in Computer Science
 - Joined Game On! - A month long game dev competition
 - Grav - Won 1st place (2012)
 - Crimson - Won 1st place mobile division (2013)